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▶ **To cite this version:**

Arshdeep Singh Sethi. SIMPLEX NUMBERS AND NUMBER THEORY: COMPLEX CONNECTIONS AND ADVANCED FORMULATIONS. 2025. ⟨hal-04949957⟩

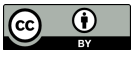
HAL Id: hal-04949957

https://hal.science/hal-04949957v1

Preprint submitted on 16 Feb 2025

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SIMPLEX NUMBERS AND NUMBER THEORY: COMPLEX CONNECTIONS AND ADVANCED FORMULATIONS

ARSHDEEP SINGH SETHI

Abstract**.** This paper introduces a novel closed-form expression for Fibonacci numbers, independent of recurrence, the golden ratio, or irrational numbers. It explores Fibonacci's connection to complex analysis, linking it to imaginary numbers and the complex plane. The study extends Fibonacci numbers to non-integer, irrational, and negative indices, providing a generalized framework. A new summation formula is derived without recurrence dependence. Additionally, Mersenne numbers are reformulated, and Simplex Numbers are introduced to derive Fibonacci terms. The paper further explores higher-dimensional structures, simplicial relations, and special-order properties, offering new insights into number theory and mathematical structures.

1. Introduction

The study of higher-dimensional simplicial complexity can be advanced by utilizing the special order of Simplex Numbers. This allows for an exploration of the rate of change in the expansion of simplicial structures as the dimensionality increases. In an n-dimensional space, an n-simplex is defined as the convex hull of n+1 affinely independent points. The simplex's structure consists of its vertices, edges, faces, and higher-dimensional analogs, all of which can be manipulated according to the combinatorial rules governing simplicial complexes. This manipulation offers insights into the evolution of a topological space by examining the ways in which simplices are connected and transformed over time. In higher-dimensional spaces, simplicial complexes provide a powerful framework for understanding the global structure of the space, facilitating a deeper investigation into the underlying topological properties.

Certain special orders of simplicial complexes can unveil profound connections with number theory, particularly in understanding how sequences evolve and interact in higher-dimensional spaces. This approach could potentially lead to the development of new formulations for existing sequences in number theory, offering fresh perspectives and insights into their underlying structures.

Studying the growth of simplicial complexes in higher dimensions, particularly with an alternating odd-even order of n-dimensional faces, provides valuable insights into the uniform expansion of n-simplices.

$$S_0^1, S_1^1, S_2^1 + S_0^2, S_3^1 + S_1^2, S_4^1 + S_2^2 + S_0^3, S_5^1 + S_3^2 + S_1^3 \dots S_n^k$$

The sequence represents the special order growth of n-dimensional structures(faces) within n-simplices in higher dimensions. Here, S_n denotes the n-dimensional structure(faces) of S^k, where S^k represents an n-simplex. The combined notation S_n^k encapsulates the hierarchical expansion of n-dimensional simplicial structures, revealing intricate patterns of growth and connectivity in higher-dimensional spaces. It is known that:

$$S_n^{k+1} = f_k = \binom{n+1}{k+1} = \frac{(n+1)!}{(k+1)!\,(n-k)!}$$

and

$$S_n^k = f_{k-1} = \binom{n+1}{k} = \frac{(n+1)!}{(k)!\,(n-(k+1))!}$$

Thus, the sequence S_n^k can be written as:

$$S_n^k = \frac{n}{1!}, \frac{n(n+1)}{2!}, \frac{n(n+1)(n+2)}{3!}, \frac{n(n+1)(n+2)(n+3)}{4!} \dots$$

This notation demonstrates the expansion of the simplicial structures as they grow in higher dimensions, with each term representing the growth of n-dimensional simplices at different stages.

By combining the simplicial complexes in an alternating odd-even n^{th} term sequence, we obtain the following expression:

$N1 = S_1^1$
$N2 = S_2^1 + S_0^2$
$N3 = S_3^1 + S_1^2$
$N4 = S_4^1 + S_2^2 + S_0^3$
$N5 = S_5^1 + S_3^2 + S_1^3$
$N6 = S_6^1 + S_4^2 + S_2^3 + S_0^4$

By systematically adding the simplices and analyzing their structural composition, we can determine the total number of faces in the simplicial complex. Given the alternating odd-

even ordering of n-dimensional faces, the cumulative count of faces can be obtained by summing the individual contributions of each simplex across all dimensions.

$N1 = 1$ =1
$N2 = 2 + 0$ =2
$N3 = 3 + 1$ =4
$N4 = 4 + 3 + 0$ =7

$N5 = 5 + 6 + 1$ =12
$N6 = 6 + 10 + 4 + 0$ =20

By applying the special order of summing within the simplicial complex, we observe an expected increase in the number of n-dimensional faces. Notably, this growth follows a pattern analogous to the Fibonacci Summation Sequence. Consequently, the total number of faces generated in an alternating odd-even n-dimensional structure aligns with the sequence of Fibonacci Summation, suggesting a deep combinatorial connection between simplicial growth and Fibonacci-like recurrence relations.

2. Fibonacci Summation Formula

Given the properties of simplex numbers and their increasing dimensions with respect to the nth term, the numerator can be expressed using the product notation as: $\prod_{i=0}^{n-1}(n+i)$, while the denominator is given by: $\prod_{n=0}^{n-1}(1+n)$, which holds true for S_n^k. Advancing this formulation by incorporating the previous conditions, we obtain: $\prod_{k=0}^{k-1}\frac{(n-2i+k)}{(1+k)}$, where S_0^k is excluded since non-integer terms are not utilized in summations. To finalize the formulation, we define i using the floor function as:

$$i = \left\lfloor \frac{n-1}{2} \right\rfloor, k = 1 + i$$

and introduce summation bounds with a lower limit of (2,1) and an upper limit of (k, i). The resulting final formula for computing the Fibonacci Summation Sequence through the nth term is:

$$\sum_{n=1}^{n} F_n = n + \sum_{(k,i)=(2,1)}^{(k,i)} \prod_{k=0}^{k-1} \frac{n-2i+k}{1+k},\ (k,\ i) = \left(i+1, \left\lfloor \frac{n-1}{2} \right\rfloor\right) \qquad (1)$$

Alternatively, in terms of S_n^k, the expression can be rewritten as:

$$\sum_{n=1}^{n} F_n = S_n^1 + \sum_{(k,i)=(2,1)}^{(k,i)} S_{n-2i}^k\ ,\ (\mathrm{k,\ i}) = \left(i+1, \left\lfloor \frac{n-1}{2} \right\rfloor\right)$$

Thus, this formulation presents a novel approach to deriving the Fibonacci Summation Sequence independently of the traditional Fibonacci Sequence and the Golden Ratio.

Illustration 1.1: Solving the Fibonacci Summation Sequence for the following nth term:

(a). n=7

(a)- (k, i) = (4,3)

$$\sum_{n=1}^{7} F_7 = 7 + \sum_{(k,i)=(2,1)}^{(4,3)} \prod_{k=0}^{k-1} \frac{7-2i+k}{1+k}$$

$$= 7 + \prod_{k=0}^{1} \frac{5+k}{1+k} + \prod_{k=0}^{2} \frac{3+k}{1+k} + \prod_{k=0}^{3} \frac{1+k}{1+k}$$

$$= 7 + \left(\frac{5}{1}\right)\left(\frac{6}{2}\right) + \left(\frac{3}{1}\right)\left(\frac{4}{2}\right)\left(\frac{5}{3}\right) + \left(\frac{1}{1}\right)\left(\frac{2}{2}\right)\left(\frac{3}{3}\right)\left(\frac{4}{4}\right)$$

$$\sum_{n=1}^{7} F_7 = 7+15+10+1 = 33$$

3. Fibonacci Formula

To derive the formula for the Fibonacci Sequence, we begin by utilizing the Fibonacci Summation Sequence. We excluded the simplex S-values with n=0 due to the inherent nature of n in the Fibonacci Summation Sequence, where n is always an integer. As a result, the (k, i) values are modified to $\left(i+1, \left\lfloor \frac{n-2}{2} \right\rfloor\right)$, ensuring the formula accommodates integer, non-integer, irrational, or negative non-integer values of n. Thus, the formula for calculating the Fibonacci Sequence through the nth term is:

$$F_n = 1 + \left[(n-2) + \sum_{(k,i)=(2,1)}^{(k,i)} \prod_{k=0}^{k-1} \frac{n-2(1+i)+k}{1+k}\right], \text{(k. i)} = \left(i+1, \left\lfloor \frac{n-2}{2} \right\rfloor\right) \qquad (2)$$

Alternatively, in terms of S_n^k, the expression can be rewritten as:

$$F_n = S^0 + \left[S_{n-2}^1 + \sum_{(k,i)=(2,1)}^{(k,i)} S_{n-2(1+i)}^k\right], \text{(k, i)} = \left(i+1, \left\lfloor \frac{n-2}{2} \right\rfloor\right)$$

This analysis provides fundamental structure of the Fibonacci Sequence, offering an alternative formulation that diverges from traditional methodologies. By establishing a connection between simplicial complex expansion and Fibonacci growth, this approach presents a novel perspective on the sequence's generation and the dynamic behavior of its expansion in n-dimensional faces.

Illustration 1.2 Solving Fibonacci Sequence for the following nth term: -

(a). n = 6.99

(a)- (k,1) = (3,2) for n = 6.99

$$F_{6.99} = 1 + \left[4.99 + \sum_{(k.i)=(2,1)}^{(3,2)} \prod_{k=0}^{k-1} \frac{6.99-2(1+i)+k}{1+k}\right]$$

$$= 1 + \left[4.99 + \prod_{k=0}^{1} \frac{2.99+k}{1+k} + \prod_{k=0}^{2} \frac{0.99+k}{1+k}\right]$$

$$= 1 + 4.99 + \left(\frac{2.99}{1}\right)\left(\frac{3.99}{2}\right) + \left(\frac{0.99}{1}\right)\left(\frac{1.99}{2}\right)\left(\frac{2.99}{3}\right)$$

$$= 1 + 4.99 + 5.96505 + 0.995$$

$$F_{6.99} = 12.95005$$

After rigorously testing the Simplex-based Fibonacci formula for large values of n, we analyzed its behaviour by graphing it for n = 100, considering both integer and non-integer inputs. The resulting plots, presented in Figure 1.1 and Figure 2.1, reveal a smooth trajectory, further validating the robustness of our derived formulation. These graphical representations illustrate how the formula maintains structural consistency across both discrete and continuous domains, offering deeper insights into its applicability beyond traditional integer-based Fibonacci computations

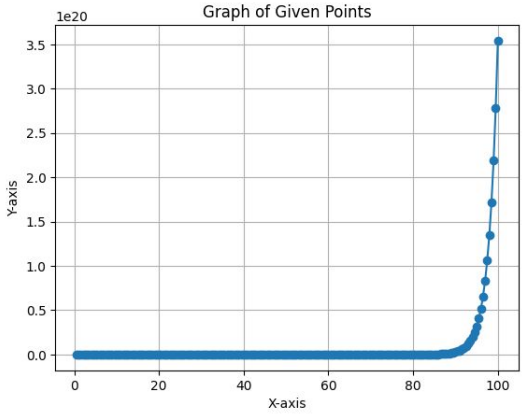


Figure 1.1

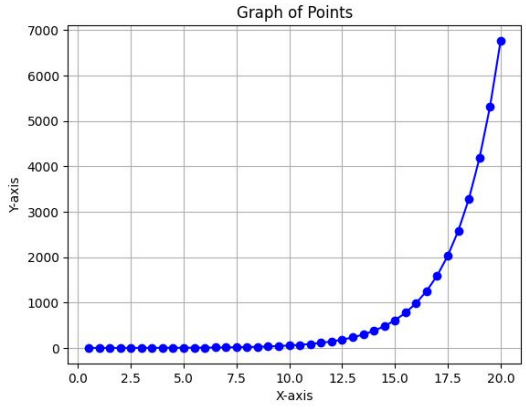


Figure 2.1

4. Negative Fibonacci Sequence

By leveraging the Fibonacci formulation, we can extend our analysis to derive its negative counterpart. This allows us to explore the inverse properties of the sequence within the context of simplicial complex expansion.

$$F_{-n} = (-1)^{n+1}\left[1 + (n-2) + \sum_{(k,i)=(2,1)}^{(k,i)} \prod_{k=0}^{k-1} \frac{n-2(1+i)+k}{1+k}\right], \text{(k. i)} = \left(i+1, \left\lfloor \frac{n-2}{2} \right\rfloor\right)$$

In the case of non-integer n, the term $(-1)^{n+1}$ inherently becomes imaginary. To account for this, we can represent it in polar form as:

$$(-1)^{n+1} = \left(e^{i\pi}\right)^{n+1}$$

This formulation ensures the expression accommodates both real and imaginary components when n is non-integer.

$$F_{-n} = \left(e^{i\pi}\right)^{n+1}\left[1 + (n-2) + \sum_{(k,i)=(2,1)}^{(k,i)} \prod_{k=0}^{k-1} \frac{n-2(1+i)+k}{1+k}\right], \text{(k. i)} = \left(i+1, \left\lfloor \frac{n-2}{2} \right\rfloor\right) \qquad (3)$$

Alternatively, in terms of S_n^k, the expression can be rewritten as:

$$F_{-n} = \left(e^{i\pi}\right)^{n+1}\left[S^0 + S_{n-2}^1 + \sum_{(k,i)=(2,1)}^{(k,i)} S_{n-2(1+i)}^k\right], \text{(k. i)} = \left(i+1, \left\lfloor \frac{n-2}{2} \right\rfloor\right)$$

Illustration 1.3 Solving Fibonacci Sequence for the following nth term: -

(a). n = -7.99

(a)- (k, i) = (3,2) for n = -7.99

$$F_{-7.99} = \left(e^{i\pi}\right)^{8.99}\left[1 + 5.99 + \sum_{(k,i)=(2,1)}^{(3,2)} \prod_{k=0}^{k-1} \frac{7.99 - 2(1+i) + k}{1+k}\right]$$

$$= \left(e^{i\pi}\right)^{8.99}\left[6.99 + \prod_{k=0}^{1}\frac{3.99+k}{1+k} + \prod_{k=0}^{2}\frac{1.99+k}{1+k}\right]$$

$$= \left(e^{i\pi}\right)^{8.99}\left[6.99 + \left(\frac{3.99}{1}\right)\left(\frac{4.99}{2}\right) + \left(\frac{1.99}{1}\right)\left(\frac{2.99}{2}\right)\left(\frac{3.99}{3}\right)\right]$$

$$= (\cos(8.99\pi) + i\sin(8.99\pi))[6.99 + 9.95505 + 3.9568165]$$

$$F_{-7.99} = 20.9018665\cos(8.99\pi) + 20.9018665i\ \sin(8.99\pi)$$

$$F_{-7.99} \approx -20.891552690638715 + 0.6565434929146828\text{i}$$

5. Mersenne Number Formulas

Extending the simplicial connection to higher dimensions while maintaining the alternating odd-even n-dimensional faces, we introduce an additional structural modification by distributing n-simplices in an even-order pattern as they expand into higher-dimensional simplices. This refined growth mechanism preserves the combinatorial complexity of the structure while revealing new patterns in the distribution of faces across dimensions.

$N1 = S_1^2$
$N2 = S_2^2 + S_0^4$
$N3 = S_3^2 + S_1^4$
$N4 = S_4^2 + S_2^4 + S_0^6$
$N5 = S_5^2 + S_3^4 + S_1^6$
$N6 = S_6^2 + S_4^4 + S_2^6 + S_0^8$

By summing over the alternating odd-even n-dimensional faces while incorporating the even-order distribution of n-simplices, we obtain the following:

$N1 = 1$ =1
$N2 = 3 + 0$ =3
$N3 = 6 + 1$ =7
$N4 = 10 + 5 + 0$ =15
$N5 = 15 + 15 + 1$ =31
$N6 = 21 + 35 + 7 + 0$ =63

As we introduce an even-order distribution of n-simplices (k), the rate of increase in n-dimensional faces exhibits further acceleration, as expected. This structural modification enhances the growth dynamics of the simplicial complex, leading to a more rapid expansion in the total number of faces. Notably, the total count of faces across increasing dimensions follows the sequence of Mersenne Numbers. This reveals a fundamental connection between simplicial growth and exponential combinatorial structures, providing

a novel perspective on the emergence of Mersenne patterns in higher-dimensional topologies.

For Mersenne numbers, we can apply the same criteria by utilizing the product and setting the (k, i) values to start from (4,1), as the second column begins with K = 4 and increases uniformly. Consequently, the Mersenne number formula for the nth term, derived with the assistance of simplex numbers, is given by:

$$M_n = \frac{n(n+1)}{2} + \sum_{(k,i)=(4,1)}^{(k,i)} \prod_{k=0}^{k-1} \frac{n-2i+k}{1+k} = (\text{k, i}) = \left(2(i+1), \left\lfloor \frac{n}{2} \right\rfloor\right) \qquad (4)$$

Alternatively, in terms of S_n^k, the expression can be rewritten as:

$$M_n = S_n^2 + \sum_{(k,i)=(4,1)}^{(k,i)} S_{n-2i}^k \text{ , (k, i)} = \left(2(i+1), \left\lfloor \frac{n}{2} \right\rfloor\right)$$

Illustration 2.1 Solving Mersenne Number Sequence for the nth term: -

(a). n = 6

(k, i) = (8,3) for n = 6

$$M_6 = 21 + \sum_{(k,i)=(4,1)}^{(k,i)} \prod_{k=0}^{k-1} \frac{6-2i+k}{1+k}$$

$$= 21 + \prod_{k=0}^{3} \frac{4+k}{1+k} + \prod_{k=0}^{5} \frac{2+k}{1+k} + \prod_{k=0}^{7} \frac{0+k}{1+k}$$

$$= 21 + \left(\frac{4}{1}\right)\left(\frac{5}{2}\right)\left(\frac{6}{3}\right)\left(\frac{7}{4}\right) + \left(\frac{2}{1}\right)\left(\frac{3}{2}\right)\left(\frac{4}{3}\right)\left(\frac{5}{4}\right)\left(\frac{6}{5}\right)\left(\frac{7}{6}\right) + 0$$

$$M_6 = 21+35+7 = 63$$

Now, utilizing the Mersenne number summation formula we can incorporate into the previously derived Mersenne number formula through simplex numbers. By adding 1 to the nth values and subtracting 2 and n from the formula, we obtain:

$$\sum_{n=1}^{n} M_n = \frac{n^2+n}{2} + \sum_{(k,i)=(4,1)}^{(k,i)} \prod_{k=0}^{k-1} \frac{n-(2i-1)+k}{1+k} \text{ , (k, i)} = \left(2(i+1), \left\lfloor \frac{n}{2} \right\rfloor\right) \qquad (5)$$

Illustration 2.2 Solving Mersenne Number Summation Sequence for nth term: -

(a). n = 3

(k, i) = (4,1) for n = 3

$$\sum_{n=1}^{3} M_3 = 6 + \sum_{(k,i)=(4,1)}^{(4,1)} \prod_{k=0}^{k-1} \frac{3-(2i-1)+k}{1+k}$$

$$= 6 + \prod_{k=0}^{3} \frac{2+k}{1+k} = 6 + \left(\frac{2}{1}\right)\left(\frac{3}{2}\right)\left(\frac{4}{3}\right)\left(\frac{5}{4}\right)$$

$$\sum_{n=1}^{3} M_3 = 6 + 5 = 11$$

6. Lucas Number Formulas

Building upon the established connection between simplicial complexes and the Fibonacci sequence, we introduce a modified growth mechanism by incorporating a doubling process. In this approach, each n-dimensional simplex is systematically doubled such that the duplicated portion of the n-dimensional face or structure is iteratively repeated within a distinct, expanding n-simplex. This structured duplication leads to a compounded growth pattern, influencing the overall combinatorial complexity and altering the expansion dynamics of the simplicial complex.

$N2 = S_1^1$
$N3 = S_2^1$
$N4 = S_3^1 + S_1^1 + S_1^2$
$N5 = S_4^1 + S_2^1 + S_2^2$
$N6 = S_5^1 + S_3^1 + S_3^2 + S_1^2 + S_1^3$
$N7 = S_6^1 + S_4^1 + S_4^2 + S_2^2 + S_2^3$

By summing over the contributions of each n-dimensional simplex under this doubling mechanism, we obtain the following:

$N2 = 1$ =1
$N3 = 2$ =2
$N4 = 3 + 1 + 1$ =5
$N5 = 4 + 2 + 3$ =9
$N6 = 5 + 3 + 6 + 1 + 1$ =16
$N7 = 6 + 4 + 10 + 3 + 4$ =27

By adding 2 to each n-dimensional face count, we obtain the sequence 3,4,7,11,18,29…, which corresponds to the Lucas Number Sequence. Utilizing the previously derived Fibonacci formulation, we can restructure the sequence by incorporating the doubling mechanism inherent in the simplicial complex growth model. This transformation allows

us to reinterpret the Lucas sequence through the lens of simplicial complex expansion, where the additional terms correspond to the systematic duplication of n-dimensional structures.

$$L_n = 2 + \left[(n-1) + \sum_{(k_1,i_1)=(2,1)}^{(k_1,i_1)} \prod_{k_1=0}^{k_1-1} \frac{n-(1+2i_1)+k_1}{1+k_1} + (n-3) + \sum_{(k_2,i_2)(2,1)}^{(k_2,i_2)} \prod_{k_2=0}^{k_2-1} \frac{n-(3+2i_2)+k_2}{1+k_2}\right], (k_1,i_1),(k_2,i_2) = \left(i_1+1, \left\lfloor \frac{n-1}{2} \right\rfloor\right), (i_1, i_1-1) \quad (6)$$

It is important to note that for values of (n-1) and (n-3), the max function is applied to ensure the result is zero, as negative values are not valid for simplex numbers. Therefore, we express this as max(n-1,0) and max(n-3,0). This condition ensures that negative indices are properly handled in the context of simplex numbers.

Illustration 3.1 Solving Lucas Number Sequence for the nth term: -

(a). n = 6

$(k_1,i_1),(k_2,i_2)$ = (3,2), (2,1)

$$L_6 = 2 + \left[8 + \sum_{(k_1,i_1)=(2,1)}^{(3,2)} \prod_{k_1=0}^{k_1-1} \frac{6-(1+2i_1)+k_1}{1+k_1} + \sum_{(k_2,i_2)=(2,1)}^{(2,1)} \prod_{k_2=0}^{k_2-1} \frac{6-(3+2i_2)+k_2}{1+k_2}\right]$$

$$= 10 + \prod_{k_1=0}^{1} \frac{3+k_1}{1+k_1} + \prod_{k_1=0}^{2} \frac{1+k_1}{1+k_1} + \prod_{k_2=0}^{1} \frac{1+k_2}{1+k_2}$$

$$= 10 + \left(\frac{3}{1}\right)\left(\frac{4}{2}\right) + \left(\frac{1}{1}\right)\left(\frac{2}{2}\right)\left(\frac{3}{3}\right) + \left(\frac{1}{1}\right)\left(\frac{2}{2}\right)$$

$$L_6 = 10 + 6 + 1 + 1 = 18$$

For the summation of the Lucas number sequence and by adjusting our Lucas number formula to account for the summation sequence, we derive the following expression:

$$\sum_{n=1}^{n} L_n = \left[(n+1) + \sum_{(k_1,i_1)=(2,1)}^{(k_1,i_1)} \prod_{k_1=0}^{k_1-1} \frac{n-(2i_1-1)+k_1}{1+k_1} + (n-1) + \sum_{(k_2,i_2)(2,1)}^{(k_2,i_2)} \prod_{k_2=0}^{k_2-1} \frac{n-(1+2i_2)+k_2}{1+k_2}\right], (k_1,i_1),(k_2,i_2) = \left(i_1+1, \left\lfloor \frac{n+1}{2} \right\rfloor\right), (i_1, i_1-1) \quad (7)$$

To accommodate negative non-integer values of n, we can express this using polar form. Specifically, for non-integer n, we have $(-1)^n = e^{i\pi n}$. Thus, the Lucas number formula for negative indices and non-integer values of n can be written as:

$$L_{-n} = e^{i\pi n}\left[2 + (n-1) + \sum_{(k_1,i_1)=(2,1)}^{(k_1,i_1)} \prod_{k_1=0}^{k_1-1} \frac{n-(1+2i_1)+k_1}{1+k_1} + (n-3) + \sum_{(k_2,i_2)(2,1)}^{(k_2,i_2)} \prod_{k_2=0}^{k_2-1} \frac{n-(3+2i_2)+k_2}{1+k_2}\right], (k_1,i_1),(k_2,i_2) = \left(i_1+1, \left\lfloor \frac{n-1}{2} \right\rfloor\right), (i_1, i_1-1) \quad (8)$$

7. Topological Aspects

The development of the Fibonacci simplicial complex follows a sequential augmentation through increasing dimensions, characterized by the following stages:

$N1 = S_1^1$
$N2 = S_2^1 + S_0^2$
$N3 = S_3^1 + S_1^2$
$N4 = S_4^1 + S_2^2 + S_0^3$
$N5 = S_5^1 + S_3^2 + S_1^3$
$N6 = S_6^1 + S_4^2 + S_2^3 + S_0^4$

Each successive stage introduces higher-dimensional simplices while maintaining topological connectivity, thereby forming a systematically expanding simplicial complex.

The Betti numbers β_k provide a rigorous characterization of the evolving topological structure:

β_0(Connected Components):

- The complex remains fully connected throughout all stages of evolution.
- Final result: $\beta_0 = 1$.

β_1(1D Homology – Loops):

- The incorporation of 2D simplices progressively eliminates independent cycles.
- Final result: $\beta_1 = 0$.

β_2(2D Homology – Cavities):

- The emergence of 3D simplices leads to the collapse of any residual 2D voids.
- Final result: $\beta_2 = 0$.

β_3(3D Homology – Voids):

- No higher-dimensional voids persist.
- Final result: $\beta_3 = 0$.

The Euler characteristic is computed using the alternating sum formula:

$$X = V - E + F - T$$
$$X = 7 - 21 + 35 - 15 = 6$$

This result implies that the structure remains topologically contractible, indicative of a simply connected space without higher-dimensional homological features.

The evolution of the complex demonstrates a progressive increase in dimensionality while preserving simple connectivity. The vanishing of all Betti numbers except β_0 confirms that the final structure behaves as a contractible n-simplex, devoid of persistent homological features. The absence of nontrivial cycles or voids supports the interpretation of the structure as a higher-dimensional contractible space. Thus, the simplicial complex exhibits a homotopy-equivalent transformation into a higher-dimensional simplex, offering a mathematically tractable model for exploring connectivity, contractibility, and homotopy properties in higher-dimensional topological spaces.

The Mersenne Number system undergoes the following transformations:

$$N1 = S_1^2$$
$$N2 = S_2^2 + S_0^4$$
$$N3 = S_3^2 + S_1^4$$
$$N4 = S_4^2 + S_2^4 + S_0^6$$
$$N5 = S_5^2 + S_3^4 + S_1^6$$
$$N6 = S_6^2 + S_4^4 + S_2^6 + S_0^8$$

The Betti numbers describe the evolving topological structure:

β_0(Connected Components):

- The simplices remain connected throughout evolution.
- Final result: $\beta_0 = 1$.

β_1(1D Homology – Loops):

- The presence of 2D simplices minimizes persistent loops.
- Final result: $\beta_1 = 0$.

β_2(2D Homology – Cavities):

- The addition of 4D simplices removes any 2D cavities.
- Final result: $\beta_2 = 0$.

β_3(3D Homology – Voids):

- Higher-dimensional simplices further contract any voids.
- Final result: $\beta_3 = 0$.

β_4(4-D Voids):

- The presence of 6D simplices ensures there are no remaining higher-dimensional voids.
- Final Result: $\beta_4 = 0$

Using the general formula for simplicial complexes:

$$X = V - E + F - T \ldots$$
$$X = 7 - 21 + 35 - 15 + 4 - 2 = 8$$

This suggests that the evolving simplicial structure remains contractible, confirming its topological triviality.

The system evolves into a higher-dimensional contractible complex. The vanishing Betti numbers (except β_0) indicate that the structure remains simply connected without persistent holes or voids. The growth pattern suggests a progressive embedding of simplices into higher dimensions, making it a useful framework for studying higher-dimensional homotopy and topological invariants. Ultimately, the evolution results in a contractible higher-dimensional simplicial complex, with topological properties analogous to a higher-dimensional simplex.

The transformation of the Lucas Number simplicial structure follows a systematic progression:

$N2 = S_1^1$
$N3 = S_2^1$
$N4 = S_3^1 + S_1^1 + S_1^2$
$N5 = S_4^1 + S_2^1 + S_2^2$
$N6 = S_5^1 + S_3^1 + S_3^2 + S_1^2 + S_1^3$
$N7 = S_6^1 + S_4^1 + S_4^2 + S_2^2 + S_2^3$

Each stage extends the dimensional complexity of the simplicial complex while maintaining a coherent topological structure, signifying an evolutionary process toward higher-dimensional embeddings.
The Betti numbers of the simplicial complex follow a Fibonacci-like structure, where $\beta_0 = 1$ and $\beta_k = 0$ for $k > 0$. This pattern indicates a single connected component and vanishing higher homology groups, aligning with the combinatorial properties of Fibonacci sequences.

The Euler characteristic is determined by:

$$X = V - E + F - T$$
$$X = 6 - 15 + 20 - 6 = 5$$

This confirms that the evolving structure remains topologically contractible while preserving fundamental invariants.

The evolution results in a higher-dimensional contractible simplicial complex. The progressive integration of simplices across dimensions aligns with higher-dimensional homotopy theory and topological invariants. Ultimately, the evolving system models a contractible higher-dimensional simplicial structure, supporting investigations into topological embeddings and their applications in complex systems.

8. Comparison

To systematically analyze the effectiveness of the Simplex-based Fibonacci formula, we compare it with existing known formulas in a structured tabular format. This comparison highlights key differences, including computational efficiency, underlying mathematical principles, and structural variations.

Criteria	Binet's Formula	Matrix Formula	Simplex Formula
Generality	No	Yes	Yes
Accuracy	Yes	Yes	Yes
Applicability to Non-Integer Values	Partially	Partially	Yes
Applicability to Negative Fibonacci Numbers	No	Yes	Yes
Applicability to Negative Non-Integer Fibonacci Numbers	No	No	Yes
Connection to Complex Numbers	No	Partially	Yes
Multidimensional Application	No	No	Yes
Closed Formula for Summation of Fibonacci Terms	No	No	Yes
Connection with Mersenne and Lucas Numbers	No	No	Yes
Works for very large n-values	Partially	Yes	Yes

Table 1.1 *Comparison of Fibonacci Formulas*

As demonstrated in Table 1, the Simplex-based Fibonacci formula unveils new mathematical insights and previously unexplored connections. This novel approach not only broadens our understanding of Fibonacci-related sequences but also establishes a

deeper link between simplex numbers and integer sequences, offering potential applications in number theory and combinatorial mathematics.

Further comparing the values of Fibonacci numbers for non-integer and negative non-integer values of n:

Nth Values	Binet's Formula	Simplex Formula	Binet's Modified Formula
π	-1.27207 + 0.27207i	2.141592653	2.11703
6.99	5.066 – 0.1175i	12.95005	12.93773
-5.5	0.2077 – 0.5561i	6.375i	0.0317
-7.99	0.0577 – 0.1582i	-20.891552690638715 + 0.6565434929146828i	-20.88878

Table 2: Comparison of Values

As shown in Table 2, Binet's standard formula does not align with the Fibonacci sequence, whereas the modified version of Binet's formula aligns well, albeit with an additional v variable in $\cos(v\pi)$. Meanwhile, the Simplex-based Fibonacci formula aligns perfectly with the Fibonacci sequence without the need for any extra variables, simply following the increase in n-dimensional faces. Both formulas track the Fibonacci sequence and its graph curve with different values. However, for negative non-integer values, the Simplex-based formula exhibits a smooth wave-like behavior in the complex plane, unlike the modified Binet's formula, which shows a wave-like behavior in the real plane.

Conclusion

This paper has explored the intricate connections between simplicial complexes and the Fibonacci sequence, offering a novel perspective on the expansion of n-dimensional simplicial structures in higher dimensions. By investigating the special order growth of simplicial faces in alternating odd-even sequences, we have derived a Fibonacci Summation Sequence that provides a fresh formulation independent of the traditional Fibonacci sequence and the Golden Ratio. The analysis has extended this framework to both integer and non-integer values of n, introducing a robust methodology for calculating Fibonacci-like sequences using simplicial numbers.

Furthermore, the paper examined the behavior of the Fibonacci sequence at both positive and negative values of n, highlighting the potential for complex number applications in the context of non-integer terms. The incorporation of polar form for negative Fibonacci terms adds a layer of versatility, making this approach suitable for broader mathematical applications. The graphical validation of the derived formulas has demonstrated their consistency across different domains, strengthening the credibility of the presented formulations.

In conclusion, the connections between simplicial complexes and Fibonacci growth provide a powerful combinatorial tool, shedding light on the structure and behavior of higher-dimensional spaces. This exploration paves the way for further research into the applications of Fibonacci-like sequences in number theory, combinatorics, and topological studies, opening new avenues for the study of mathematical structures in higher dimensions.

Acknowledgements

I would like to sincerely thank my cousin, Gurjot Singh, for coding my formulas and testing them with large n values, which led to successful results. His assistance was crucial in bringing this work to fruition.

I sincerely thank Professor Rupam Barman and Dr. H. Laxmi of IIT Guwahati for helping me complete my endorsement process on arXiv, and enabling me to share my research.

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